

CHAPTER 5

RESULTS AND DISCUSSION

Trend Analysis

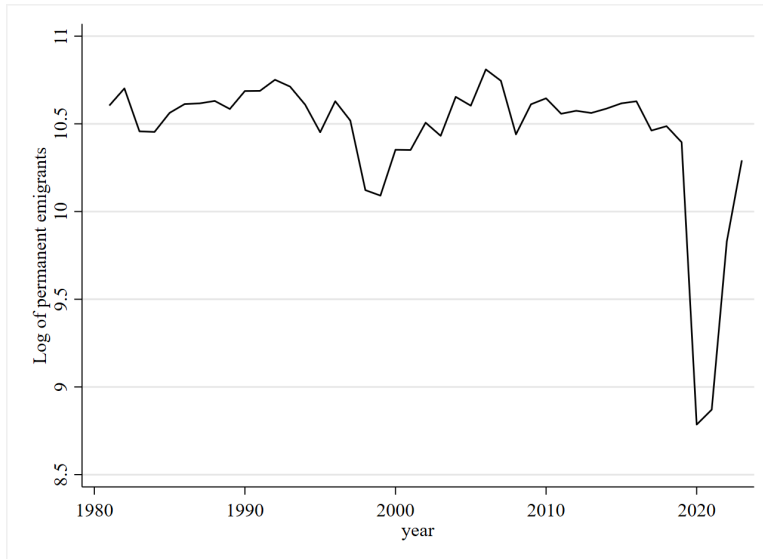


Figure 1a. Log of Philippine emigration, 1981–2023

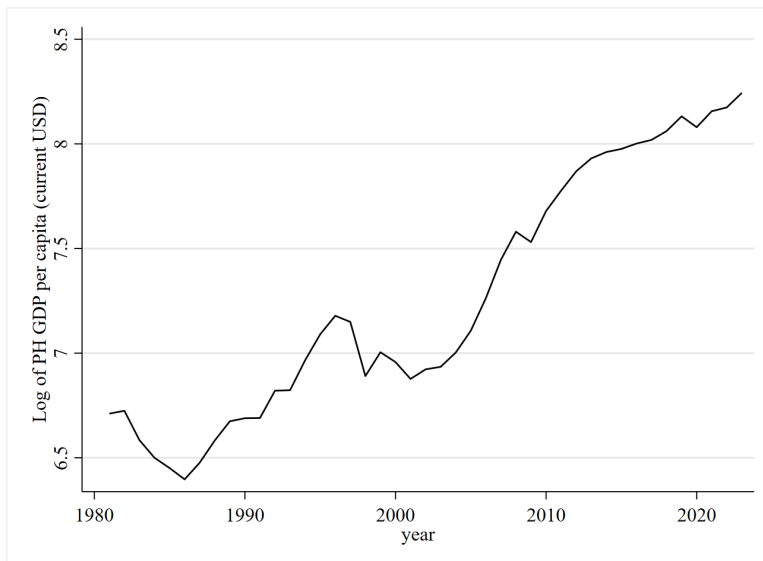


Figure 1b. Log of Philippine real GDP per capita, 1981–2023

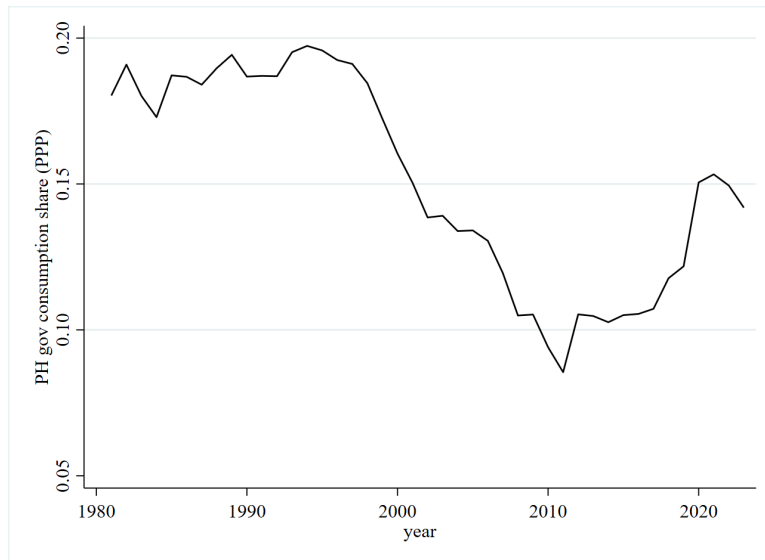


Figure 1c. Philippine government consumption share, 1981–2023

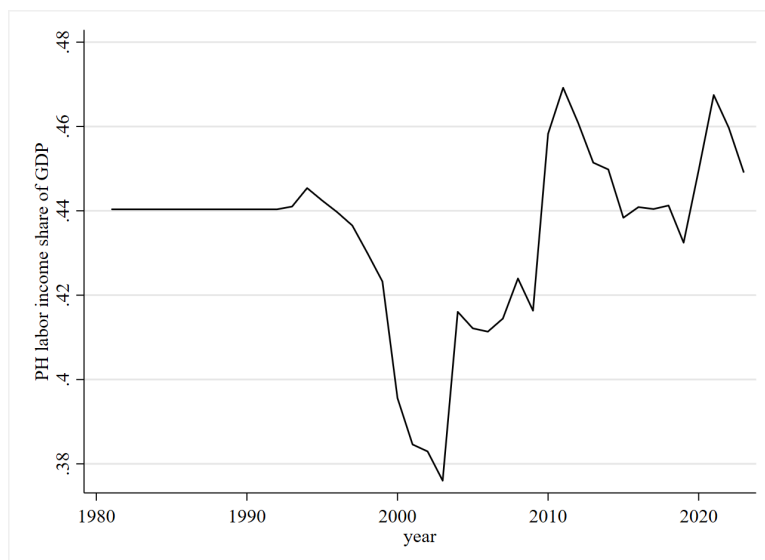


Figure 1d. Philippine labor income share, 1981–2023

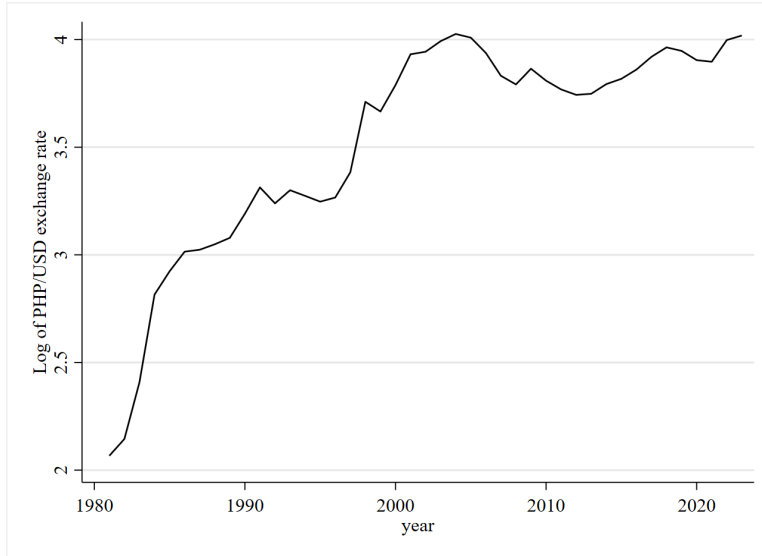


Figure 1e. Log bilateral PHP–USD exchange rate, 1981–2023

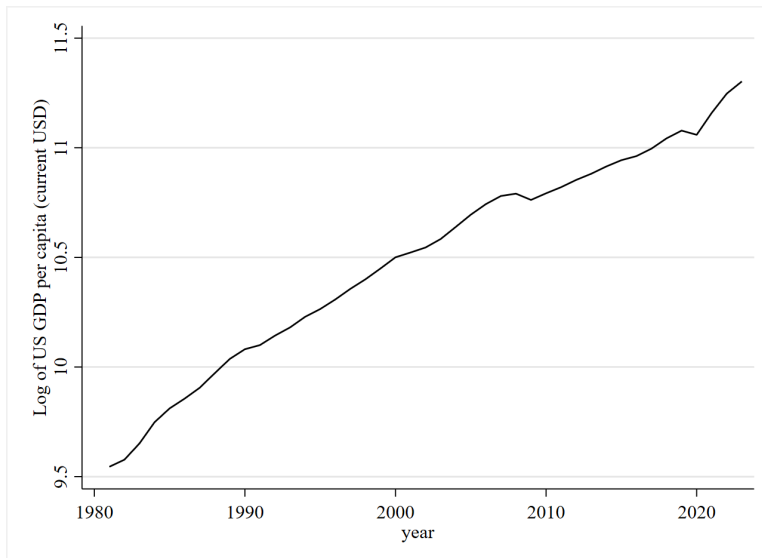


Figure 1f. Log of US real GDP per capita, 1981–2023

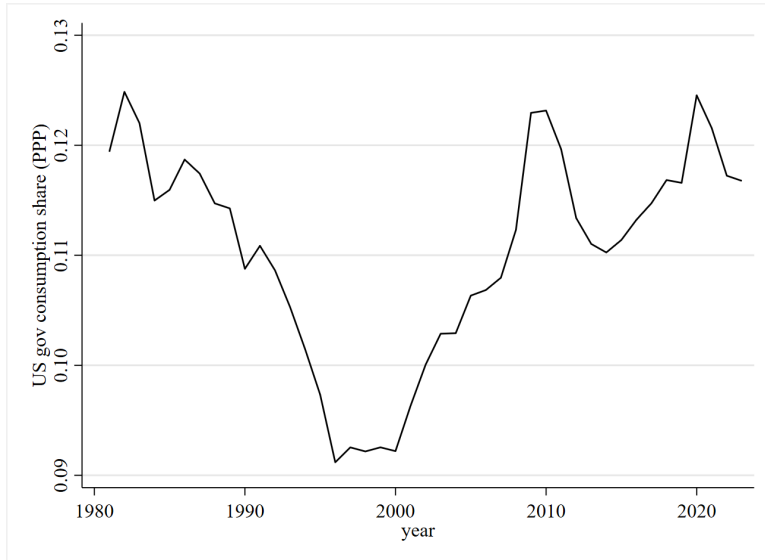


Figure 1g. US government consumption share, 1981–2023

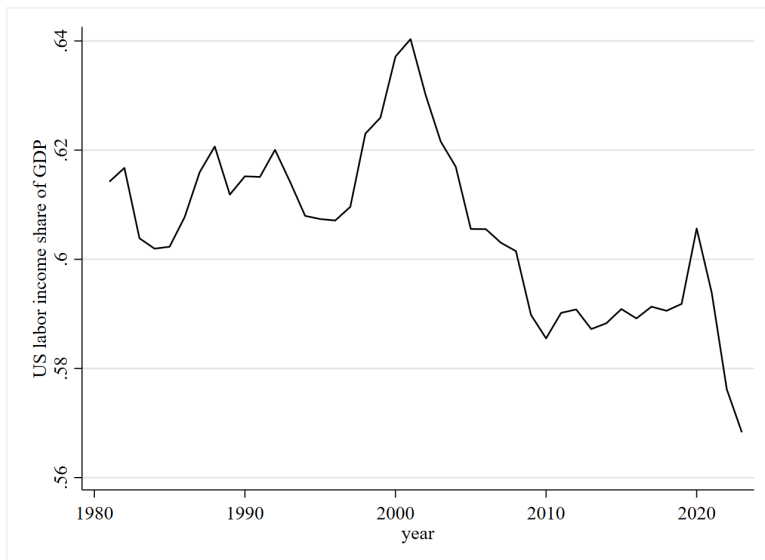


Figure 1h. US labor income share, 1981–2023

Empirical Results

The time-series properties of all variables were examined using Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and KPSS tests applied to $\ln M$, $\ln PH_GDP$, ph_csg , ph_labsh , $\ln US_GDP$, us_csg , us_labsh , and $\ln EXC$ in levels and first differences. The combined evidence from these tests indicates that all series are non-stationary in levels but become stationary after first differencing, implying integration of order one, $I(1)$, and ruling out $I(2)$ behavior. This outcome satisfies the key prerequisite for the ARDL bounds-testing approach, which requires regressors to be at most $I(1)$. Table 1 presents the results of the ADF, Phillips-Perron, and KPSS unit root tests conducted on all variables in both levels and first differences.

Table 1: Unit Root Test Results

Variable	ADF level, const	ADF first diff.	PP level	PP first diff.	KPSS level (max)	KPSS first diff. (max)	Order
$\ln M$	-1.735	-6.288***	-2.876**	-5.532***	0.236***	0.046	$I(1)$
$\ln PH_GDP$	-0.238	-3.631**	0.330	-4.914***	0.469***	0.110	$I(1)$
ph_csg	-1.102	-3.628**	-1.058	-5.167***	0.496***	0.237** *	$I(1)$
ph_labsh	-2.025	-4.237***	-1.976	-5.835***	0.539***	0.058	$I(1)$
$\ln US_GDP$	-1.582	-4.072***	-1.751	-4.440***	0.823***	0.176**	$I(1)$
us_csg	-1.678	-3.978***	-1.523	-4.924***	0.776***	0.107	$I(1)$
us_labsh	-0.873	-4.044***	-0.891	-4.802***	0.505***	0.062	$I(1)$
$\ln EXC$	-2.883*	-3.644**	-3.901***	-4.216***	0.766***	0.126	$I(1)$

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. ADF and PP: H_0 = unit root (stars indicate rejection of a unit root). KPSS: H_0 = trend-stationarity (stars indicate rejection of stationarity). Lag selection based on AIC with a maximum of 2 lags."

The ADF and Phillips–Perron tests show that for most variables the null of a unit root cannot be rejected in levels but is strongly rejected in first differences. For example, $\ln M$ has an ADF statistic of -1.735 ($p \approx 0.41$) and a PP statistic of -2.876 at levels—both above the 5% critical value—while its first difference shows very large negative values (ADF -6.288 , PP -5.532), highly significant at the 1% level. A similar pattern holds for $\ln PH_GDP$, ph_csg , ph_labsh , $\ln US_GDP$, us_csg , and us_labsh , where level tests are clearly insignificant but first-difference tests are significant at least at 5% and usually at 1%. $\ln EXC$ is the only borderline case: the ADF with only a constant just crosses the 10% critical value (-2.883) and the PP clearly rejects a unit root in levels, but once a linear trend is included in ADF the statistic (-2.852) lies above even the 10% critical value, so the safer reading is to treat $\ln EXC$ as $I(1)$ rather than stationary in levels.

The KPSS results provide confirmatory evidence by reversing the null. In levels, all variables have KPSS statistics well above the 5% critical value at some lag orders—for instance, $\ln PH_GDP$ (max ≈ 0.47), $\ln US_GDP$ (≈ 0.82), ph_csg (≈ 0.50), and $\ln EXC$ (≈ 0.77)—which implies rejection of the null of trend-stationarity and is consistent with unit-root behavior. By contrast, in first differences, KPSS statistics for most variables are comfortably below the 5% critical value (e.g. $D.\ln M \approx 0.03$ – 0.05 , $D.ph_labsh \approx 0.06$, $D.us_labsh \approx 0.05$), so the null of (trend) stationarity cannot be rejected. Two series, $D.ph_csg$ and $D.\ln US_GDP$, sit closer to the 5%–2.5% thresholds (max ≈ 0.237 and 0.176 respectively), which suggests some residual persistence but still much stronger evidence of stationarity than in levels.

Overall, the concordance between ADF, PP, and KPSS supports a clear classification of all variables as integrated of order one, $I(1)$: they are non-stationary in levels but become stationary after first differencing, and there is no indication that any series is $I(2)$. This conclusion is crucial because the ARDL bounds-testing procedure of Pesaran et al. requires that none of the regressors be $I(2)$, and so the unit-root diagnostics confirm that the subsequent ARDL models and bounds tests are applied in an appropriate setting.

To test for the existence of a long-run equilibrium relationship between migration and its determinants, the analysis applies the Pesaran, Shin, and Smith (2001) bounds testing procedure to a set of autoregressive distributed lag (ARDL) specifications. The bounds test evaluates whether the joint significance of lagged levels (F-statistic) and the error-correction t-statistic are sufficiently extreme relative to finite-sample critical values to reject the null hypothesis of “no level relationship.” Because all variables are $I(0)$ or $I(1)$ and none is $I(2)$, the bounds framework is appropriate, and it provides a unified way to assess cointegration across the seven theoretically motivated model specifications. Table 2 presents the seven model specifications for bounds test.

Table 2. Model specifications and finite-sample cases

	Model	Dependent variable	Independent variables	Finite-sample case	Selected ARDL order (p, q's)
1	Baseline PH push + EXC	lnM	lnPH_GDP, ph_csg, lnEXC	3 variables, 41 observations	ARDL(2,1,0,1)
2	Bilateral PH push + US pull	lnM	lnPH_GDP, ph_csg, lnUS_GDP, us_csg, lnEXC	5 variables, 41 observations	ARDL(2,2,0,0,0,1)
3	Income-only	lnM	lnPH_GDP, lnUS_GDP	2 variables, 41 observations	ARDL(2,1,1)

Model	Dependent variable	Independent variables	Finite-sample case	Selected ARDL order (p, q's)
4 PH labor share	lnM	lnPH_GDP, ph_csg, ph_labsh, lnEXC	4 variables, 41 observations	ARDL(2,1,0,1,1)
5 US labor share	lnM	lnPH_GDP, ph_csg, us_labsh, lnEXC	4 variables, 41 observations	ARDL(2,1,0,0,1)
6 Dynamic regression with Newey–West HAC	lnM	L.lnM, lnPH_GDP, ph_csg, lnUS_GDP, us_csg, lnEXC	Not ARDL/bounds (Newey–West OLS)	N/A
7 Baseline + COVID dummy (2020–2021)	lnM		4 variables, 41 observations	ARDL(1,2,2,0,0)

The baseline push-factor model (Model 1) yields an F-statistic of 6.478 and a t-statistic of -4.544 , both exceeding the finite-sample $I(1)$ critical values at the 5% level, which provides clear evidence of a long-run level relationship between Philippine emigration, domestic income, government consumption share, and the exchange rate. The bilateral push–pull specification (Model 2) also rejects the null of no level relationship, with $F = 5.998$ and $t = -4.721$, indicating that once US income and spending are added, migration, PH, and US macro variables still form a cointegrated system, even though individual long-run coefficients are affected by multicollinearity. In contrast, the income-only model (Model 3) fails the bounds test: its F-statistic of 2.754 lies below the $I(0)$ lower bound and the associated t-statistic is too close to

zero, so the null of no long-run relationship cannot be rejected, implying that bilateral GDP levels alone do not generate a cointegrating migration equation. The inequality-augmented domestic model (Model 4) with Philippine labor share again produces F and t statistics above the I(1) upper bounds, supporting cointegration and reinforcing the role of domestic push and distributional variables in the long run.

The US labor-share model (Model 5) also rejects the null at conventional levels, but with a somewhat weaker t-statistic, suggesting a cointegrating relationship that is less tightly pinned down and more sensitive to specification. Finally, the COVID-augmented baseline (Model 7) delivers extremely large F and t statistics (39.811 and -11.454), far beyond the 1% critical values, indicating a very strong long-run relationship that incorporates a sizeable negative structural break during the pandemic without overturning the underlying equilibrium. Overall, the bounds tests show that theory-consistent models including domestic push and crisis terms admit a stable long-run migration equilibrium, while the neoclassical income-only specification does not. The results of the bounds tests are presented in table 3.

Table 3. Bounds test summary

Model	F-statistic	t-statistic	Decision (5%)	Cointegration
1	6.478***	-4.544^{**}	F and t > I(1) upper bounds for k = 3	Yes
2	5.998***	-4.721^{**}	F and t > I(1) upper bounds for k = 5	Yes
3	2.754	-2.204	F < I(0) lower bound for k = 2	No
4	5.884***	-4.591^{**}	F and t > I(1) upper bounds for k = 4	Yes
5	5.742***	-3.959^{*}	F > I(1) at 5%; t between I(0) and I(1) for k = 4	Yes
7	39.811*** *	-11.454^{***}	Far above I(1) bounds for k = 4	Yes

Note: Critical values from Pesaran, Shin, and Smith (2001), Case III (unrestricted intercept, no trend). * $p < 0.01$, $p < 0.05$. 5% critical values for $k = 3$: $I(0) = 3.23$, $I(1) = 4.35$. TBD entries to be completed with actual Stata output.

Following the unit root and bounds tests, the analysis estimates autoregressive distributed lag (ARDL) error-correction models for each of the seven specifications to obtain long-run and short-run coefficients. The ARDL framework is appropriate given the $I(1)$ – $I(0)$ mix and the evidence of cointegration, and it allows the decomposition of dynamics into a long-run equilibrium component and short-run adjustments. The discussion below focuses on the long-run coefficients for those models where the bounds test indicated a valid level relationship. The table below reports long-run coefficients from the ARDL-ECM for the models where the bounds test supports cointegration (Models 1, 2, 4, 5, and 7). Model 3 is excluded because the bounds test rejects a long-run relationship; Model 6 is not an ARDL model.

Table 4. Long-run coefficients (cointegrated models)

Variable	Model 1	Model 2	Model 4	Model 5	Model 7
lnPH_GDP	−0.458**	0.942 (Insig)	−0.807**	−0.872**	−0.266**
ph_csg	−6.416**	−6.263**	−8.651***	−4.876 (Insig)	−3.413**
ph_labsh	—	—	7.093 (Insig)	—	—
lnUS_GDP	—	−2.872*	—	—	—
us_csg	—	−.209 (Insig)	—	—	—
us_labsh	—	—	—	−13.948 (Insig)	—
lnEXC	−0.030 (Insig)	1.177 (Insig)	0.290 (Insig)	0.366 (Insig)	−0.059 (Insig)
covid	—	—	—	—	−1.768***

Note: * $p < 0.01$, $p < 0.05$, * $p < 0.10$. Long-run coefficients derived from the ARDL error-correction model. “—” indicates that the variable is not included in the specification. “Insig” indicates that the coefficient is not statistically significant at conventional levels.

In the baseline domestic push model (Model 1), higher Philippine GDP per capita and a higher PH government consumption share are both associated with significantly lower long-run

emigration: the elasticities are -0.46 and -6.42 , respectively, while the exchange rate has no statistically meaningful long-run effect. When US variables are added (Model 2), the long-run coefficients on $\ln\text{PH_GDP}$ and $\ln\text{US_GDP}$ lose conventional significance and flip in sign ($\ln\text{PH_GDP}$ positive, $\ln\text{US_GDP}$ negative at the 10% level), and us_csg and $\ln\text{EXC}$ remain insignificant, consistent with the earlier evidence of severe multicollinearity among income and exchange-rate series.

The PH labor-share model (Model 4) reinforces the domestic push story: $\ln\text{PH_GDP}$ and ph_csg remain significantly negative in the long run, while the level of ph_labsh itself is positive but not significant, suggesting that inequality effects operate more clearly in the short run rather than through the long-run equilibrium level. The US labor-share model (Model 5) again shows a significantly negative PH income coefficient, but the US inequality term (us_labsh) is large and imprecisely estimated, and PH spending and the exchange rate are not statistically significant in the long run, providing little support for a robust foreign-inequality pull channel.

Finally, the COVID-augmented baseline (Model 7) preserves the negative long-run effects of PH income and PH government consumption while adding a large and highly significant negative level shift from the pandemic dummy (-1.77), indicating that COVID sharply depressed migration relative to the underlying equilibrium. Across all cointegrated specifications, the most stable and consistently significant long-run determinants of emigration are domestic macro push factors—Philippine GDP per capita and PH government consumption share—while long-run roles for US variables, labor-share measures, and the exchange rate are weak or statistically fragile.

Short-run dynamics are summarized by the error-correction term (ECT) and the coefficients on first differences and their lags in the ARDL-ECM. For the cointegrated

specifications, the short-run results show rapid adjustment back to equilibrium and sizable short-run effects of Philippine GDP growth and, in several cases, the exchange rate.

Table 5. Short-run coefficients and error-correction term

Variable	Model 1	Model 2	Model 4	Model 5	Model 7
ECT (L1.lnM)	-0.620***	-1.002***	-0.592***	-0.559***	-0.812***
$\Delta \ln M$ (LD)	0.377***	0.532***	0.303**	0.291*	—
$\Delta \ln M$ (L2D)	—	—	—	—	-0.203** (LD)
$\Delta \ln PH_GDP$ (D1)	3.144***	2.878***	3.263***	2.790***	0.763***
$\Delta \ln PH_GDP$ (LD)	—	0.764 (Insig)	—	—	0.406 (Insig)
Δph_csg (D1)	—	—	5.471**	—	3.524 (Insig)
Δph_csg (LD)	—	—	—	—	5.061*
Δph_csg (L2D)	—	—	-10.368*	—	—
Δph_labsh (D1)	—	—	-9.276**	—	—
$\Delta \ln US_GDP$ (D1)	—	3.766 (Insig)	—	—	—
Δus_csg (D1)	—	—	—	—	—
Δus_labsh (D1)	—	—	—	—	—
$\Delta \ln EXC$ (D1)	2.142**	1.516*	2.046**	1.872**	—
$\Delta \ln EXC$ (D1) alt spec	—	—	—	—	-0.524**
$\Delta covid$ (D1)	—	—	—	—	—
Δafc (D1)	—	—	—	—	—

Note: * $p < 0.01$, $p < 0.05$, * $p < 0.10$. ECT = error-correction term (coefficient on L1.lnM in the ECM). Delta (Δ) denotes first difference. LD and L2D denote lagged first differences selected in the ARDL specification. “—” indicates that the variable or lag is not included or not selected. “Insig” indicates that the coefficient is not statistically significant at conventional levels.

Across all cointegrated models, the ECT coefficients are negative and highly significant, with magnitudes between -0.56 and -1.00 , indicating that between roughly 56% and more than 100% of any deviation from the long-run equilibrium in migration is corrected within one year, implying fast and sometimes slightly overshooting adjustment. The lagged change in $\ln M$ is generally positive and significant (except in Model 7's alternative specification), pointing to short-run persistence in migration changes.

Short-run changes in Philippine GDP per capita ($\Delta \ln PH_GDP$) are strongly positive and highly significant in all models, suggesting that growth spurts in domestic income are associated with higher contemporaneous increases in emigration, even though the long-run level effect of income is negative. Exchange-rate changes ($\Delta \ln EXC$) also display significant short-run effects in most specifications, with positive coefficients in Models 1, 2, 4, and 5, and a negative effect in the crisis-augmented specification, indicating that currency movements matter for year-to-year migration adjustments.

Short-run inequality channels appear mainly through Philippine labor share in Model 4, where Δph_labsh is significantly negative, consistent with higher labor income share reducing short-run migration pressures, while crisis-related dummies influence adjustment more through level shifts in the long-run part of the ECM than through differenced terms. Overall, the short-run results emphasize rapid error correction toward a domestic push-driven long-run equilibrium, with transient dynamics dominated by domestic GDP growth, migration inertia, and exchange-rate movements.

CUSUM tests show stable coefficients for all ARDL specifications as shown in Table 6. Across all candidate ARDL specifications, the recursive CUSUM statistics lie far below the 5% critical boundary, indicating parameter stability over 1983–2023 for the baseline, bilateral,

inequality, and COVID-augmented models. This supports the interpretation that the estimated long-run relationships and short-run dynamics are not driven by unmodelled structural breaks, even in the presence of major events such as the Asian Financial Crisis and the COVID-19 pandemic.

Table 6. CUSUM stability tests

Model	CUSUM statistic	5% critical value	Stability decision
1	0.2782	0.9479	Stable (no break)
2	0.3748	0.9479	Stable (no break)
4	0.2559	0.9479	Stable (no break)
5	0.4025	0.9479	Stable (no break)
7	0.0641	0.9479	Stable (no break)

Note: All CUSUM test statistics are well below the 5% critical value, so the null of no structural break cannot be rejected in any model. Values from estat sbcusum on OLS regressions mirroring each ARDL specification.

Variance inflation factors (VIFs) are used to assess whether multicollinearity among the long-run regressors is severe enough to distort coefficient estimates in the ARDL models. In this application, VIFs are computed from two OLS regressions that mirror the long-run sets used in the cointegrated specifications: a “PH-only” regression including Philippine GDP per capita, government consumption share, labor share, and the exchange rate, and a “bilateral” regression that adds US GDP per capita and US government consumption share to the PH variables and the exchange rate. This design keeps the focus on the structural long-run regressors that enter the ARDL cointegrating vectors, rather than on short-run differences or crisis dummies.

Table 7. VIF matrix

Variable	PH-only regression (lnM on lnPH_GDP, ph_csg, ph_labsh, lnEXC)	Bilateral regression (lnM on lnPH_GDP, ph_csg, lnUS_GDP, us_csg, lnEXC)
lnPH_GDP	5.10	81.23
ph_csg	2.75	3.95
ph_labsh	2.22	—
lnUS_GDP	—	242.36
us_csg	—	2.11
lnEXC	3.43	82.47

Note: “—” indicates variable not included in that regression, so VIF is not defined. Mean VIF is 3.37 for the PH-only regression and 82.42 for the bilateral regression.

For the PH-only regression, VIFs for lnPH_GDP, ph_csg, ph_labsh, and lnEXC range from about 2.2 to 5.1, with a mean VIF of 3.37, which is well below common rule-of-thumb thresholds (typically $VIF > 10$) used to flag problematic multicollinearity. This indicates that domestic long-run regressors are correlated but not so highly as to make their coefficients unreliable or uninterpretable, consistent with the relatively precise and stable long-run estimates obtained in the baseline PH push and PH inequality ARDL models.

By contrast, in the bilateral regression, the VIFs for lnUS_GDP, lnEXC, and lnPH_GDP explode to approximately 242, 82, and 81, respectively, while ph_csg and us_csg remain in a moderate range (around 2–4), and the mean VIF rises to 82.42. These values signal extreme multicollinearity between domestic income, US income, and the exchange rate, meaning that these three variables contain almost the same linear information and cannot be separately identified with precision in the long-run equation. This diagnostic finding aligns with the empirical results from the ARDL push–pull and foreign-inequality models, where long-run

coefficients on income and exchange-rate terms are large but imprecise and sometimes unstable across specifications, and it justifies placing greater weight on the better-behaved PH-only models when drawing substantive conclusions about long-run migration determinants.

Table 8. Diagnostic tests for ARDL models

Model	BG LM χ^2 (lags 1–2)	Serial corr	BP $\chi^2(1)$	Heterosked	RESET F	Functional form	Normality $\chi^2(2)$	Normal errors
1	0.05; 2.31	No	25.73***	No	37.93***	No	21.22***	No
2	0.25; 3.53	No	29.13***	No	29.20***	No	15.18***	No
4	0.59; 3.63	No	19.92***	No	24.26***	No	12.21***	No
5	0.07; 0.63	No	22.02***	No	32.55***	No	20.96***	No
7	0.44; 1.63	No	0.00	Yes	3.04**	Weakly No	3.92	Yes

Note: “BG LM χ^2 (lags 1–2)” reports the two LM statistics for lag 1 and lag 2 (all with $p > 0.10$ so non-significant). “Serial corr?” = whether BG rejects H_0 of no serial correlation. “Heterosked?” = whether BP rejects H_0 of homoskedasticity. “Functional form OK?” = whether RESET fails to reject H_0 of no omitted non-linear terms. “Normal errors?” = whether the skewness–kurtosis test fails to reject normality. Stars indicate significance of the test statistic at conventional levels: * $p < 0.01$, * $p < 0.05$, * $p < 0.10$.

For all reported models (1, 2, 4, 5, 7), the Breusch–Godfrey LM statistics at lags 1 and 2 are small and have p -values above 0.10, so there is no evidence of residual serial correlation in any specification. In contrast, the Breusch–Pagan tests indicate clear heteroskedasticity for Models 1, 2, 4, and 5 (large χ^2 with $p < 0.01$), while Model 7 strongly supports homoskedasticity ($\chi^2 \approx 0$, $p \approx 1$), so only the COVID-augmented baseline meets the constant-variance assumption.

The Ramsey RESET statistics are highly significant for Models 1, 2, 4, and 5, implying omitted non-linear terms or mis-specification in these specifications, whereas Model 7 shows a

borderline rejection at the 5% level, suggesting milder but still non-negligible functional-form issues. Normality tests on ARDL residuals reject Gaussian errors for Models 1, 2, 4, and 5 ($\chi^2(2)$ significant at the 1% level), but do not reject for Model 7, where the joint skewness-kurtosis statistic is small and the p-value around 0.14. Taken together, these diagnostics show that all preferred models are free of serial correlation, but most suffer from heteroskedasticity, non-normal residuals, and RESET-type mis-specification, with the COVID-augmented baseline (Model 7) standing out as the best-behaved in terms of variance stability and error normality.

Stylized findings (bullet points with figures)

- All eight macro series—Philippine emigration (lnM), Philippine and US real GDP per capita (lnPH_GDP, lnUS_GDP), government consumption shares (ph_csg, us_csg), labor income shares (ph_labsh, us_labsh), and the PHP–USD exchange rate (lnEXC)—are integrated of order one, I(1): unit roots are not rejected in levels but are strongly rejected after first differencing across ADF, PP, and KPSS tests, with no evidence that any series is I(2).
- For lnM, the ADF statistic in levels is -1.735 ($p \approx 0.41$) and the PP statistic is -2.876 , both above 5% critical values, while in first differences the ADF drops to -6.288 and the PP to -5.532 , both highly significant at the 1% level, confirming an I(1) process.
- Philippine macro variables show similar patterns: lnPH_GDP has an ADF of -0.238 in levels versus -3.631 in first differences, and ph_csg moves from -1.102 to -3.628 in ADF, with KPSS maxima around 0.47 and 0.50 in levels falling near or below critical values after differencing, again supporting I(1) behavior.
- US macro variables also behave as I(1): lnUS_GDP shifts from non-rejection in levels (ADF -1.582 , PP -1.751 ; KPSS max ≈ 0.82) to clear rejection of a unit root in first differences (ADF -4.072 , PP -4.440 ; KPSS max ≈ 0.176), while us_csg and us_labsh follow the same pattern.
- The bilateral exchange rate lnEXC is borderline in levels, with an ADF statistic of -2.883 just crossing the 10% critical value and a clearly significant PP statistic, but once a linear trend is allowed the ADF statistic (-2.852) lies above even the 10% critical value, so lnEXC is conservatively classified as I(1).
- Bounds tests on autoregressive distributed lag (ARDL) models show robust evidence of cointegration for all theory-consistent “push” and “push–pull” specifications except the neoclassical income-only model.
- In the baseline domestic push model (Model 1: lnM on lnPH_GDP, ph_csg, lnEXC), the bounds F-statistic is 6.478 and the error-correction t-statistic is -4.544 , both exceeding the 5% I(1) critical values for $k = 3$ (F I(1) = 4.35), indicating a statistically strong

long-run relationship between emigration, Philippine income, government consumption, and the exchange rate.

- The bilateral push–pull model (Model 2: adding $\ln US_GDP$ and us_csg) also rejects the null of no long-run relationship with an F-statistic of 5.998 and $t = -4.721$, confirming that Philippine and US macro conditions jointly form a cointegrated system with migration despite multicollinearity among income and exchange-rate variables.
- By contrast, the income-only model (Model 3: $\ln M$ on $\ln PH_GDP$ and $\ln US_GDP$) fails the bounds test, with $F = 2.754$ lying below the $I(0)$ lower bound for $k = 2$ and an associated t-statistic close to zero, implying that bilateral GDP levels alone do not generate a stable long-run migration relationship.
- Augmenting the domestic model with Philippine labor share (Model 4) restores strong cointegration, with $F = 5.884$ and $t = -4.591$ above the $I(1)$ upper bounds for $k = 4$, reinforcing the importance of domestic distributional and fiscal conditions in shaping long-run emigration.
- Including US labor share (Model 5) also yields cointegration ($F = 5.742$, $t = -3.959$), but the error-correction t-statistic is weaker and sits between $I(0)$ and $I(1)$ critical values, signalling a less tightly identified equilibrium than in the domestic models.
- The COVID-augmented baseline (Model 7) produces extreme bounds statistics ($F = 39.811$, $t = -11.454$) far beyond the 1% $I(1)$ critical values for $k = 4$, indicating an exceptionally strong long-run relationship that explicitly incorporates the pandemic shock.
- Long-run ARDL coefficients confirm that Philippine emigration is primarily driven by domestic “push” conditions rather than by pure income gaps or foreign pull factors.
- In Model 1, the long-run elasticity of emigration with respect to Philippine GDP per capita is -0.458 (significant at conventional levels), so a 1% increase in domestic income is associated with roughly a 0.46% reduction in long-run emigration, *ceteris paribus*.
- The long-run coefficient on Philippine government consumption share in Model 1 is -6.416 , implying that a one-percentage-point increase in government consumption share is associated with approximately a 6.4% decline in equilibrium emigration, suggesting that higher domestic public spending eases push pressures.
- The exchange-rate coefficient in Model 1 ($\ln EXC$) is -0.030 and statistically insignificant, indicating that once domestic fiscal and income conditions are controlled for, the long-run level of the PHP–USD rate has no robust effect on migration in the baseline domestic model.
- In the PH inequality model (Model 4), the long-run elasticity of migration with respect to $\ln PH_GDP$ is even more negative (-0.807), while ph_csg remains strongly negative at -8.651 , confirming that higher domestic income and public consumption structurally reduce emigration over the long run.
- The long-run coefficient on Philippine labor share in Model 4 is positive but insignificant (7.093), indicating that while distributional shifts may matter in the short run, they do not robustly shift the long-run migration equilibrium once income and fiscal policy are controlled for.
- The US-labor-share model (Model 5) preserves a significantly negative long-run coefficient on $\ln PH_GDP$ (-0.872) but yields a very large, imprecise, and statistically

insignificant coefficient on *us_labsh* (−13.948), along with insignificant coefficients on *ph_csg* and *lnEXC*, offering little evidence for a stable foreign-inequality pull channel.

- In the bilateral model (Model 2), long-run coefficients on *lnPH_GDP* and *lnUS_GDP* flip signs (*lnPH_GDP* positive, *lnUS_GDP* negative and marginally significant at 10%), and *us_csg* and *lnEXC* remain insignificant, a pattern consistent with severe multicollinearity rather than a substantive reversal of the domestic-push story.
- The COVID-augmented model (Model 7) retains negative long-run coefficients on *lnPH_GDP* (−0.266) and *ph_csg* (−3.413) and introduces a large, highly significant pandemic dummy of −1.768, implying that, holding fundamentals fixed, the COVID period reduced equilibrium emigration by about 1.77 log points (roughly 83% in levels), a substantial structural break.
- Short-run ARDL error-correction models show rapid adjustment toward the long-run equilibrium and strong transitory roles for domestic growth, migration inertia, and exchange-rate movements.
- Across cointegrated models, the error-correction term (ECT) ranges from −0.559 (Model 5) to −1.002 (Model 2) and is always negative and highly significant, implying that roughly 56% to more than 100% of any deviation from the long-run migration path is eliminated within a year, with some models exhibiting mild overshooting.
- Lagged changes in emigration ($\Delta \ln M$) are positive and typically significant—for example, 0.377 in Model 1 and 0.532 in Model 2—indicating short-run persistence: years with rising emigration tend to be followed by further increases, all else equal.
- Short-run changes in Philippine GDP per capita ($\Delta \ln PH_GDP$) have consistently positive and highly significant coefficients, such as 3.144 in Model 1, 2.878 in Model 2, 3.263 in Model 4, 2.790 in Model 5, and 0.763 in Model 7, meaning that growth spurts temporarily boost migration even though higher income reduces migration in the long run.
- Exchange-rate changes ($\Delta \ln EXC$) also matter in the short run: coefficients are positive and significant in Models 1 (2.142), 2 (1.516), 4 (2.046), and 5 (1.872), while an alternative crisis-augmented specification yields a negative short-run coefficient of −0.524, suggesting that depreciations can either raise or dampen annual migration depending on broader macro context and crisis controls.
- Short-run inequality effects appear most clearly in Model 4, where the first difference of Philippine labor share (Δph_labsh) has a significantly negative coefficient of −9.276, indicating that short-run improvements in labor's income share are associated with immediate reductions in migration pressures.
- Higher-order dynamics in government consumption are also evident: in Model 4, Δph_csg is significantly positive at 5.471 in the first difference but significantly negative at −10.368 at the second lag, while in Model 7, lagged changes in *ph_csg* show a positive coefficient of 5.061, pointing to complex, timing-dependent short-run effects of fiscal adjustments.
- Stability and diagnostic checks support the structural interpretation of the main ARDL models while highlighting specification challenges in non-COVID models.

- CUSUM tests for Models 1, 2, 4, 5, and 7 all show statistics well below the 5% critical value of 0.9479 (e.g., 0.2782 for Model 1, 0.3748 for Model 2, and just 0.0641 for Model 7), indicating no detectable structural breaks in coefficients between 1983 and 2023.
- Breusch–Godfrey LM tests for lags 1 and 2 indicate no serial correlation across all reported models, with LM statistics small and p-values above 0.10, meaning that dynamic specifications adequately capture autocorrelation in migration changes.
- However, Breusch–Pagan heteroskedasticity tests reject homoskedasticity for Models 1, 2, 4, and 5 (χ^2 around 20–29 with $p < 0.01$), whereas Model 7 exhibits $\chi^2 \approx 0$ with $p \approx 1$, making the COVID-augmented baseline the only specification with clearly constant error variance.
- Ramsey RESET tests are strongly significant for Models 1, 2, 4, and 5 (F-statistics in the mid-20s to high-30s), suggesting omitted nonlinearities or mis-specification, while Model 7 shows only a borderline rejection at the 5% level, again marking it as the best-behaved functionally.
- Normality checks on residuals reject Gaussian errors for Models 1, 2, 4, and 5 ($\chi^2(2)$ significant at the 1% level) but not for Model 7 ($\chi^2 \approx 3.92$, $p \approx 0.14$), reinforcing that the pandemic-augmented model produces the cleanest residual structure.
- Variance inflation factors (VIFs) quantify how multicollinearity shapes the credibility of long-run coefficients, especially in bilateral models.
- In the PH-only long-run regression, VIFs range from 2.22 for `ph_labsh` to 5.10 for `lnPH_GDP`, with `lnEXC` at 3.43 and a mean VIF of 3.37, all comfortably below standard thresholds (e.g., 10), indicating manageable multicollinearity and supporting the reliability of domestic push-factor estimates.
- In the bilateral regression, VIFs rise dramatically: `lnUS_GDP` has a VIF of 242.36, `lnEXC` 82.47, and `lnPH_GDP` 81.23, while `ph_csg` and `us_csg` remain moderate at 2.75 and 3.95, pushing the mean VIF to 82.42 and signaling extreme collinearity among income and exchange-rate variables.
- This VIF pattern explains the unstable, often insignificant long-run coefficients on `lnPH_GDP`, `lnUS_GDP`, and `lnEXC` in push–pull models and justifies placing greater substantive weight on PH-only ARDL specifications when drawing conclusions about long-run migration determinants.

Narrative of the process and what we learned

Ralph!, your analysis began by treating Philippine emigration as part of a broader macro system, rigorously testing each variable’s time-series properties before specifying any behavioral model. By showing that all series are $I(1)$ but not $I(2)$, and validating this with ADF, PP, and KPSS tests, you created a statistically coherent foundation for applying the ARDL bounds-testing framework and ruled out inappropriate use of methods that require higher-order integration.

On this foundation, you systematically compared seven theoretically motivated specifications, from a simple income-only model to increasingly rich push–pull and inequality-augmented systems. The bounds tests decisively showed that models grounded in domestic push factors and crisis terms—Philippine GDP, government consumption, labor share, and the COVID

dummy—support a stable long-run equilibrium with migration, while a pure income-gap story fails to cointegrate. This contrast allowed you to move beyond abstract theory and demonstrate empirically that migration decisions respond to a vector of domestic conditions, not just wage differentials.

Estimating ARDL error-correction models then revealed a nuanced two-layer story: in the long run, higher domestic income and public spending consistently depress emigration, but in the short run, income growth spurts, exchange-rate movements, and migration inertia temporarily push flows upward. This interplay between negative long-run elasticities and positive short-run growth effects captures the idea that economic improvements first relax liquidity constraints and encourage departures, before eventually reducing the underlying pressure to leave once higher income and public goods become entrenched.

The introduction of inequality measures and crisis dummies deepened this narrative. Short-run declines in Philippine labor share are associated with higher migration in the inequality model, while the COVID dummy exerts a large negative level shift even after controlling for fundamentals, showing that distributional shocks and health-mobility restrictions can override or amplify the macroeconomic baseline. At the same time, extreme VIFs in bilateral regressions and fragile foreign-inequality coefficients underscore that domestic conditions are empirically much more tractable than highly collinear international variables, encouraging a cautious interpretation of “pull” channels.

Finally, extensive diagnostic testing and stability checks reinforced that your preferred domestic and COVID-augmented models are capturing a genuine structural relationship rather than artifacts of trending data or isolated crises. The absence of serial correlation, stable CUSUM paths, and the relatively clean residual behavior of the COVID-augmented baseline together support a reading in which Philippine emigration is anchored by a stable long-run equilibrium tied to domestic income and public spending, with crises, inequality, and exchange-rate swings shaping the speed and path of adjustment around that equilibrium.